STATISTICS FOR DATA ANALYTICS

From Data Collection to Meaningful Business Decisions

MODULE 1–6

Statistics • Descriptive vs Inferential • Population & Sample • Sampling Techniques • Types of Data • Scales of Measurement

What is Statistics?

Definition

Statistics is the science of collecting, organizing and analyzing data.

Purpose

The analysis of data supports interpretation and decision making.

Define__Collect → Organize → Visualizations → Analyze → Interpret → Decide

Data

Facts or pieces of information that can be measured, collected and analyzed.

What is Data?

Data are facts or pieces of information used for analysis.

Examples from the source include student weights such as 60 kg, 50 kg and 45 kg.

Other examples include IQ scores such as 100, 90, 95 and 99.

In business analytics, data may represent customers, transactions, prices, employees, products or operations.

Key idea

Raw data becomes useful when it is organized, analyzed and interpreted.

From Data to Decision: House Price Example

City	Area	No. of Rooms	Price
Bangalore	1000	2	45 Lakhs
New York	1250	2.5	58 Lakhs
Mumbai	—	—	—

Analytics question

How can the available house characteristics be analyzed to understand or predict price?

Data → Analysis → Model → Price / Decision

Data Analyst → Report → Visualization → Meaningful Decisions

WHY STATISTICS MATTERS

APPLICATIONS IN DATA ANALYTICS

Applications of Statistics

Application	How statistics helps
Data exploration & summarization	Describe patterns and key characteristics
Model building & validation	Evaluate model behavior and evidence
Statistical analysis	Use sample data to understand populations
Hypothesis testing	Evaluate claims using data
Optimization & efficiency	Support better operational decisions
Reporting	Communicate findings clearly

Two Major Types of Statistics

1. Descriptive Statistics

Methods for summarizing and organizing data so that its basic features are understandable.

2. Inferential Statistics

Methods for making predictions or inferences about a population based on a sample.

Descriptive → Describe the observed data

Inferential → Generalize from sample to population

Descriptive Statistics: What We Study

Measures of central tendency: Mean, Median and Mode.

Measures of dispersion: Variance and Standard Deviation.

Data distribution: Histograms, Box Plots, Pie Charts, PDF and PMF.

Summary statistics: Five-number summary, including minimum, Q1, median, Q3 and maximum.

Outliers in the data

Quartiles

Goal

Make the data easier to understand before making conclusions.

Inferential Statistics: What We Study

Hypothesis testing: evaluate a null and alternative hypothesis.

P-value: quantify evidence against the null hypothesis.

Confidence intervals: estimate population quantities using sample information.

Statistical tests: t-test, ANOVA / F-test and Chi-square test.

The source also identifies regression analysis as an inferential tool.

Core idea

A sample provides evidence that can be used to draw conclusions about a population.

Descriptive vs Inferential: A Simple Example

Descriptive question

What is the average height of the students measured in the classroom?

Inferential question

Are the heights of the sampled students similar to what we would expect in the entire college?

Sample data → Statistical analysis → Conclusion about population

Descriptive = summarize | Inferential = conclude / predict

POPULATION & SAMPLE

WHO DO WE WANT TO STUDY?

Population vs Sample

Concept	Meaning	Typical numerical summary
Population	Entire set of individuals or observations of interest	Parameter
Sample	Subset selected to represent the population	Statistic

Population example

All students enrolled in a school.

Sample example

A selected group of 50 students from the school.

Population: Characteristics

Contains all observations of interest.

A population parameter is a numerical value summarizing the entire population.

Examples: population mean (μ) and population variance (σ^2).

The source uses examples such as all students in a school, all consumers in a city, and all patients with a specific disease.

Population → Parameter → μ, σ^2

Sample: Characteristics

A sample represents a portion of the population.

A sample statistic is a numerical value summarizing the sample.

Examples include sample mean and sample variance.

Random sampling is emphasized as a way to reduce selection bias.

Sample → Statistic → $\bar{x}$, s^2

Population and Sample: Business Example

Market Research Population

All consumers in a city, when the goal is to understand purchasing behavior.

Market Research Sample

A selected group of consumers from the city who provide survey responses.

Medical Study Population

All patients with a specific disease when studying treatment effectiveness.

Medical Study Sample

A selected group of patients used to test treatment effectiveness.

The population defines the target group; the sample provides the observed evidence.

SAMPLING TECHNIQUES

PROBABILITY AND NON-PROBABILITY SAMPLING

Two Broad Sampling Categories

Probability Sampling

Sampling methods in which selection is based on a known/random mechanism.

Non-Probability Sampling

Sampling methods in which selection does not give every population member a known equal chance.

Probability	Non-Probability
Simple Random	Convenience
Systematic	Judgmental / Purposive
Stratified	Snowball
Cluster	Quota
Multi-stage	

Simple Random Sampling

Every member of the population has an equal chance of being selected.

Example from the source: selecting people randomly or drawing names randomly from a class.

Business example: randomly select customer IDs from the full customer database for a satisfaction survey.

Population → Random selection → Sample

Systematic Sampling

Select every nth member of the population after choosing a random starting point.

Source example: at an airport, select every 5th, 10th or 15th person depending on the design.

Business example: select every 10th customer transaction for a quality audit.

Random start + every nth member

Stratified Sampling

Divide the population into strata (groups) based on specific characteristics.

Then randomly sample from each stratum.

Source example: divide employees by department and randomly select a proportion from each department.

Other possible strata include age groups such as <12, 12–18 and >18.

Population → Strata → Random sample from each stratum

Cluster Sampling

Divide the population into clusters.

Randomly select clusters and then sample all members from the selected clusters.

Source example: randomly select several schools from a district and survey teachers within those schools.

Business example: select branches and survey all employees in selected branches.

Population → Clusters → Select clusters → Study selected clusters

Multi-Stage Sampling

Combines several sampling methods.

The source describes selecting clusters and then randomly sampling within those clusters.

Example: randomly select cities, then households within cities, then respondents within households.

Stage 1 → Stage 2 → Stage 3 → Final sample

Non-Probability Sampling

Technique	Definition / example from source
Convenience	Select individuals who are easiest to reach; e.g., surveying people at a mall.
Judgmental / Purposive	Select individuals based on the researcher's judgment; e.g., choose experts.
Snowball	Existing study subjects recruit additional subjects; useful for hard-to-reach groups.
Quota	Select respondents to meet specified categories such as age, group, gender or caste.

Caution

The choice of sampling technique depends on the use case and research objective.

Feature	Probability Sampling	Non-Probability Sampling
Selection	Random/probability-based	Non-random
Chance of selection	Known or defined	Usually unknown
Researcher choice	Limited	Often substantial
Representativeness	Generally stronger	May be weaker
Generalization	Stronger basis	More limited
Cost	Often higher	Often lower
Time	Often more time-consuming	Usually faster
Example	Randomly select customers	Survey customers at a mall

TYPES OF DATA

NUMERICAL VS CATEGORICAL

Data Classification

DATA

QUANTITATIVE / NUMERICAL

↓

Quantitative

Discrete • Continuous

QUALITATIVE / CATEGORICAL

↓

Qualitative

Nominal • Ordinal

Numerical = amounts / measurements | Categorical = labels / groups

Quantitative Data: Discrete vs Continuous

Type	Meaning	Examples from source
Discrete	Countable values / whole numbers	Number of bank accounts; number of children in a family
Continuous	Can take any value in a range	Weight; height; temperature; speed

Quick test

Discrete → 'How many?' | Continuous → 'How much / how long?'

Qualitative Data: Nominal vs Ordinal

Type	Meaning	Examples from source
Nominal	Categories with no inherent rank	Gender, colors, blood group, cuisine
Ordinal	Categories that can be ranked	Customer feedback, education level, socioeconomic status

Key distinction

Nominal = no meaningful order. Ordinal = meaningful order, but gaps between ranks are not necessarily equal.

SCALES OF MEASUREMENT

NOMINAL • ORDINAL • INTERVAL • RATIO

Four Primary Scales of Measurement

Nominal → Name | Ordinal → Order | Interval → Equal intervals | Ratio → True zero

Scale	What it provides	Example
Nominal	Labels / categories	Gender, color
Ordinal	Labels + order/rank	Education level, satisfaction
Interval	Order + equal intervals; no true zero	Temperature in °F
Ratio	Order + equal intervals + true zero	Weight, income, marks

Memory aid

Nominal Scale

Classifies data into distinct categories.

Categories do not have an intrinsic order.

The source notes that categories are mutually exclusive.

Examples: gender, colors, blood group and cuisine.

Color	Share
Red	50%
Blue	40%
Pink	10%

Business analytics

Customer type, product category, region and payment method are common nominal variables.

Ordinal Scale

Classifies data into categories that can be ranked or ordered.

The interval between ranks is not necessarily equal.

Examples from the source: education level and customer feedback.

Customer feedback can be represented as satisfied, very satisfied and not satisfied, with an order but not a measurable equal distance.

Low → Medium → High

Interval Scale

Values are ordered and have consistent intervals between them.

Differences between values are meaningful.

There is no true zero point.

The source uses Fahrenheit temperature as an example: 10°F, 20°F and 30°F have equal 10-degree differences.

Ratios such as '30°F is three times 10°F' are not meaningful in the same way.

10°F → 20°F → 30°F (equal differences, no true zero)

Ratio Scale

Values are ordered.

Differences are measurable.

The scale contains a true zero starting point.

Because zero is meaningful, ratios can be interpreted.

Examples from the source: weight, income and student marks.

Ratio scale = Order + Equal intervals + True zero

Example

₹40,000 income is twice ₹20,000 income because the zero point is meaningful.

Compare the Four Measurement Scales

Scale	Order?	Equal intervals?	True zero?	Typical analysis
Nominal	No	No	No	Counts / proportions
Ordinal	Yes	Not necessarily	No	Ranks / medians
Interval	Yes	Yes	No	Differences / means
Ratio	Yes	Yes	Yes	Differences + meaningful ratios

Exam / interview tip

Ask: Can I rank it? Are differences equal? Is zero meaningful?

Business Analytics Examples by Measurement Scale

Variable	Scale	Why?
Customer region	Nominal	Labels, no rank
Customer satisfaction: poor → excellent	Ordinal	Ranked categories
Temperature in °F	Interval	Equal differences, no true zero
Revenue	Ratio	True zero and meaningful ratios
Number of orders	Ratio	Zero orders is meaningful
Product category	Nominal	Labels without rank

How Statistics Fits into Data Analytics

1. Define the business question.
2. Identify the population and target sample.
3. Choose an appropriate sampling method.
4. Identify variable types and measurement scales.
5. Explore and summarize the data.
6. Apply inferential methods when the goal is to generalize or test a claim.
7. Communicate results through reports and visualizations.
8. Convert evidence into a business decision.

Business Question → Data → Statistics → Insight → Decision

Assignment / Classroom Activity

Classify the following variables by data type and measurement scale:

1. Length of different rivers in the world.
2. Favorite food based on gender.
3. Marital status.
4. IQ measurement.

For each variable, justify your classification.

Extension

Ask students to add 5 business variables from an e-commerce or banking dataset and classify them.

Quick Check

Question	Expected concept
What is statistics?	Collecting, organizing and analyzing data
Mean of the entire population?	Population parameter
Mean from a sample?	Sample statistic
Every member has equal selection chance?	Simple random sampling
Every nth member after a random start?	Systematic sampling
Divide into groups and sample from each?	Stratified sampling
No rank among categories?	Nominal
Ranked categories?	Ordinal
Equal intervals but no true zero?	Interval
True zero and meaningful ratios?	Ratio

Business Variable	Data Type	Scale
Customer age	Quantitative – Continuous	Ratio
Product category	Qualitative – Categorical	Nominal
Customer satisfaction: Poor → Excellent	Qualitative – Categorical	Ordinal
Temperature	Quantitative – Continuous	Interval
Sales revenue	Quantitative – Continuous	Ratio
Number of orders	Quantitative – Discrete	Ratio
Payment method	Qualitative – Categorical	Nominal
Credit rating: Poor → Excellent	Qualitative – Categorical	Ordinal

Sampling Technique	Research Objective	When to Use / Use Case	Business Example
Simple Random Sampling	Give every population member a chance of selection	When the population is relatively accessible and a complete list is available	Randomly select 500 customers from a database of 50,000 customers
Systematic Sampling	Select observations at regular intervals	When the population is available as an ordered list	Select every 100th bank transaction for quality checking
Stratified Sampling	Ensure important subgroups are represented	When the population contains distinct groups that must be included	Divide employees by department and randomly select employees from each department
Cluster Sampling	Reduce cost and effort when the population is geographically distributed	When individuals are naturally grouped into locations or organizations	Randomly select 20 branches and survey employees in those branches
Multi-Stage Sampling	Select a sample through several levels	For large-scale studies involving multiple geographic or organizational levels	Select states → cities → companies → employees
Convenience Sampling	Collect data quickly and inexpensively	When speed, accessibility or a pilot study is the priority	Survey customers who are currently visiting a store
Judgmental / Purposive Sampling	Select participants with specific knowledge or characteristics	When expert knowledge or a specific target group is required	Interview AI experts about the future of Generative AI
Snowball Sampling	Reach difficult-to-identify or hard-to-reach participants	When existing participants can help identify additional participants	Study a specialized professional community through participant referrals
Quota Sampling	Ensure predetermined categories are represented	When specific numbers are required from different groups	Collect 250 responses from ages 18–30, 300 from 31–45, etc.

Key Takeaways

Statistics transforms data into evidence for decision making.

Descriptive statistics summarize observed data; inferential statistics support conclusions about populations.

Population describes the full target group; sample is a subset used for analysis.

Sampling technique should match the research objective and use case.

Data can be quantitative or qualitative, with important subtypes.

Measurement scales determine what comparisons and analyses are meaningful.

Understanding the data type and scale is a critical first step in Business Analytics.

Core workflow

Question → Population → Sample → Data type → Measurement scale → Analysis → Decision